

AI Action Plan – for the classroom

Response to Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Matt Kanjirath

Application of AI tools for student learning

With the use of Artificial Intelligence, we can focus on using technology to understand and customize learning to an individual's needs through the clicks and responses made by each student. We could determine if the student is color blind, autistic, left brain, or right brain and customize the curriculum to the student's needs. I recommend that the classroom teacher be made the business owner of the class, use the laptop to understand the child, and provide human interaction where necessary. The curriculum can be loaded onto the laptop from the cloud and modified by an authorized central authority, creating an environment for standardized learning. The recommended approach will reduce the education hierarchy's bureaucracy, which places the teacher and staff at the bottom of the decision tree in student education.

The current US education system emphasizes standardizing testing for all students, forcing teachers to “teach to the test,” which limits creativity and flexibility in lesson planning. Therefore, teachers are forced to teach to the test, creating an environment of standardization that does not encourage diverse learning styles of special needs and gifted students in the classroom.

Public schools, especially in low-income areas, suffer from insufficient funding, outdated materials, and large class sizes. Teachers assigned to large class sizes have high administrative burdens and excessive paperwork, which results in undue stress. Socioeconomic factors, such as food insecurity and unstable home environments, impact student performance and cannot adequately addressed by schools. Schools often prioritize academic subjects over essential life skills like financial literacy, mental health, and vocational training. High college tuition costs create obstacles for many students pursuing higher education.

The contemporary United States education system significantly emphasizes standardized testing for all students, compelling educators to focus primarily on test preparation. This approach restricts creativity and flexibility in lesson planning, ultimately hindering the ability to accommodate diverse learning styles, particularly for special needs and gifted students.

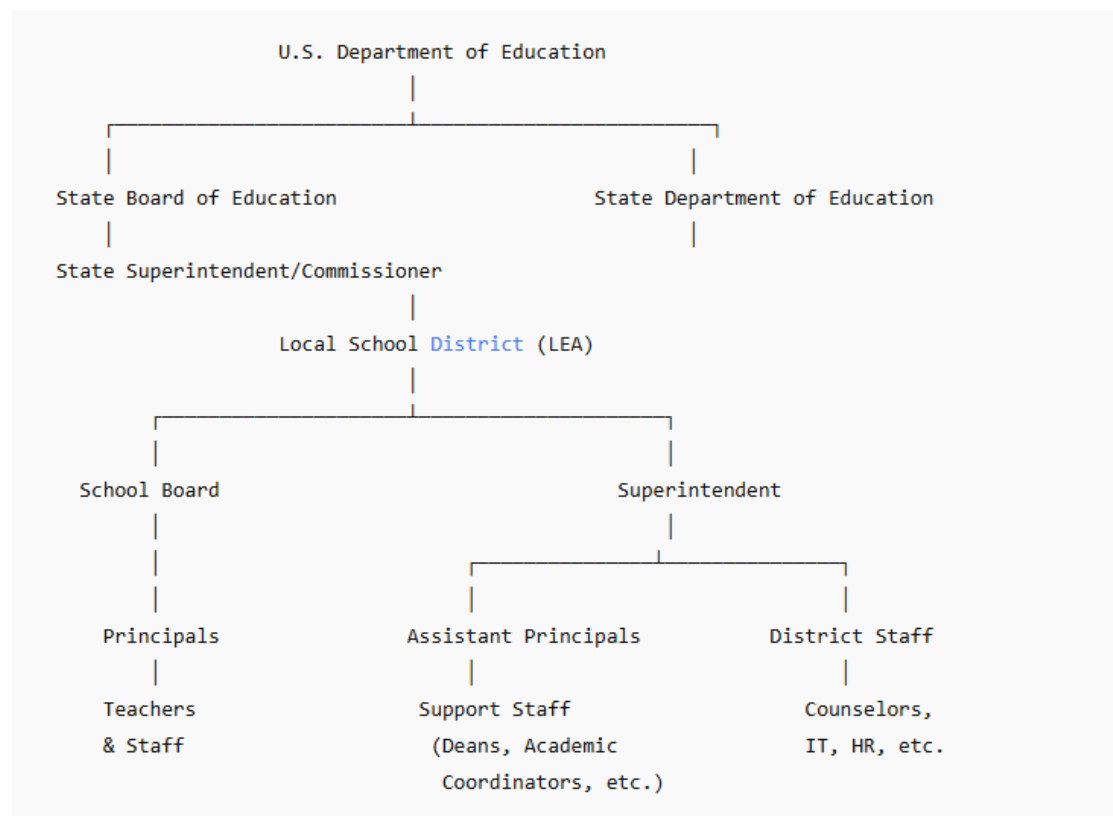
Public schools, particularly those in low-income areas, face challenges such as inadequate funding, outdated resources, and large class sizes. Educators managing large classrooms encounter substantial administrative burdens and excessive paperwork, leading to increased stress levels. Socioeconomic factors, including food insecurity and unstable home environments, adversely affect student performance and are often inadequately addressed by educational institutions. Furthermore, schools frequently prioritize academic subjects at the expense of essential life skills, such as financial literacy, mental health awareness, and vocational training. The rising college tuition costs present additional barriers for many students seeking higher education.

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Recognizing that a classroom teacher functions similarly to a business owner is essential. Teachers need to be provided with the autonomy and flexibility to tailor their teaching using Artificial Intelligence to understand and meet the specific needs of their students. Within a broad framework, they could achieve more effective educational outcomes. The hierarchical structure within educational institutions places teachers and staff at the lowest tier of decision-making authority. We first need to understand that a class teacher is like the business owner of the class, and given the independence and freedom to teach according to needs with a broad guideline, they could apply to get the end result. The current hierarchy positions place the teachers and staff at the lowest decision-making ladder.



Recommendation

To empower educators and school staff, delegating the responsibility and authority for implementing teaching standards to Artificial Intelligence (AI) is essential. AI possesses the transformative potential to enhance student learning by personalizing educational experiences, increasing engagement, and optimizing administrative processes. One of AI's most significant advantages is its capacity to customize learning experiences based on individual student needs. AI can assess students' strengths and weaknesses through adaptive learning platforms, offering

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tailored resources and exercises that align with their unique learning pace and style. This personalized approach facilitates a deeper comprehension of complex concepts.

A good AI-augmented learning model for school-going children should balance personalization, engagement, social interaction, and teacher involvement. Below is an example of a structured approach that integrates AI while maintaining essential human guidance.

Hybrid Learning Model (AI + Human Teachers)

The best approach is a blended learning model, in which AI personalizes lessons while teachers provide supervision, emotional support, and deeper engagement. Key Components:

1. **Personalized Learning Paths (AI-driven)**
AI adapts content based on individual student's strengths, weaknesses, and learning pace.
2. **Interactive & Gamified Learning (Engagement)**
 - AI-powered games, quizzes, and simulations make learning fun and improve retention.
 - Example: AI-based coding games like CodeCombat.
3. **Collaborative AI-Assisted Group Learning**
 - AI facilitates peer learning by grouping students based on skill levels and interests.
 - Virtual AI mentors guide group projects, discussion forums, or debates.
4. **Real-time Feedback & Assessment**
 - AI tracks student progress and provides instant feedback on mistakes.
 - Teachers receive detailed reports on student performance, allowing targeted intervention.
5. **Adaptive AI Tutors & Chatbots**
 - AI-powered tutors answer student queries in real-time, reducing their dependency on teachers for routine questions.
 - Example: Google's Read Along AI helps children with reading fluency.
6. **AI-Powered Creativity Tools**
 - AI tools assist in storytelling, art, and music, encouraging creativity.
 - Example: DALL·E or AI-powered storytelling apps generate interactive narratives based on prompts.
7. **Ethical & Safe AI Use**
 - AI models must be bias-free, explainable, and privacy-compliant to protect children.
 - Schools should ensure data is securely managed and parents have transparency in AI decision-making.

A voice response system can be an alternative to chatbots or other technologies for students with disabilities, ensuring that assistance is readily available for those in need.

AI can support educators by automating administrative tasks such as grading and attendance tracking. This automation allows teachers to concentrate more on instruction and student interaction, ultimately enhancing the quality of education. AI can also analyze data to identify

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trends in student performance, empowering educators to make informed decisions regarding curriculum adjustments and intervention strategies.

Implementation

With the advancements in cloud technology and data center connectivity, establishing a centralized system to develop curricula for each grade is feasible. Assigned administrators and classroom teachers can make recommendations for approval, ensuring alignment with standards set by a central authority. This initiative will streamline existing bureaucratic processes and allow for customization tailored to each grade level. Additionally, an AI tool should be developed to assess each student's learning style—whether it be right-brain, left-brain, graphic representation, or color adjustments for those who are color blind—while providing detailed examples in specific lessons as needed.

Summary

In summary, AI has the potential to revolutionize student learning in schools by personalizing education, enhancing engagement, and supporting educators. This would ultimately lead to a more effective and enriching educational experience for all students.